DSA3050A (BUSINESS INTELLIGENCE AND DATA VISUALIZATION): ADVANCED POWERBI EXAM

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1. Introduction

1.1. Domain selected

The domain I selected was ‘Retail’. This is because retail generates high-volume transactional data making it ideal for this advanced Power BI project.

1.2. Business Problem

Scenario

I have been engaged as a Business Intelligence Analyst to design an advanced Power BI solution for an online retail company (TATA) operating across multiple countries. The company sells a wide variety of consumer products and records transaction-level sales data. Senior management wants a data-driven analytical dashboard to understand performance, identify trends and risks, and uncover growth opportunities.

Problem Statement

Despite having large volumes of transactional sales data, management lacks an analytical view of sales performance across products, customers, time periods, and countries. Senior management is unable to easily identify sales trends, top-performing products, high-value customers, or regions contributing most to revenue, which limits strategic planning and performance monitoring.

1.3. Business Objectives

The Power BI solution aims to support senior management by identifying the following:

- How are sales and revenue trending over time?
- Which products and product groups perform best?
- Which countries generate the highest revenue?
- Who are the most valuable customers?
- How do returns and cancellations affect overall performance?
- Are there seasonal patterns or growth opportunities?

2. Dataset Description

2.1. Dataset selected

Dataset name: ‘TATA: Online Retail Dataset’

Source: Kaggle

Source Link: https://www.kaggle.com/datasets/ishanshrivastava28/tata-online-retail-dataset/data?select=Online+Retail+Data+Set.csv

2.2. Dataset information

This is dataset contains transactional retail data where each row represents a single product line item within an invoice. It contains both sales and returns (negative quantities).

2.3. Key Tables and Fields.

The dataset is originally a single table, but I shall later transform it into a relational star schema.

Key columns in the dataset

- InvoiceNo – identifies each transaction
- InvoiceDate – timestamp of purchase
- StockCode – unique product identifier
- Description – product name
- Quantity – number of units sold (or returned)
- UnitPrice – price per unit
- CustomerID – customer identifier
- Country – customer's country

2.4. Why this dataset is suitable for advanced power BI analysis?

This is because this dataset 'TATA: Online Retail Dataset' meets all the data requirements:

- Has more than 9,000 transaction rows
- Has categorical, and numerical fields.
- Supports star schema modeling (has enough fields to create multiple logical tables)
- Has date field which can be used for time intelligence

2.5. Screenshots

Below is a screenshot of the data source

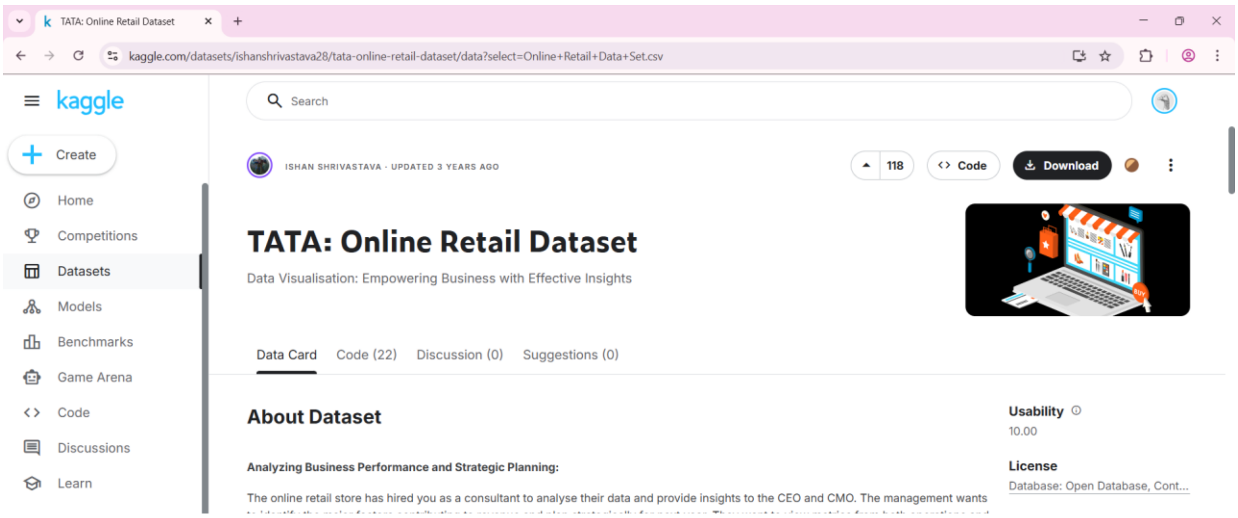


Figure 1: Screenshot of the data source

Below is a screenshot of proof of download

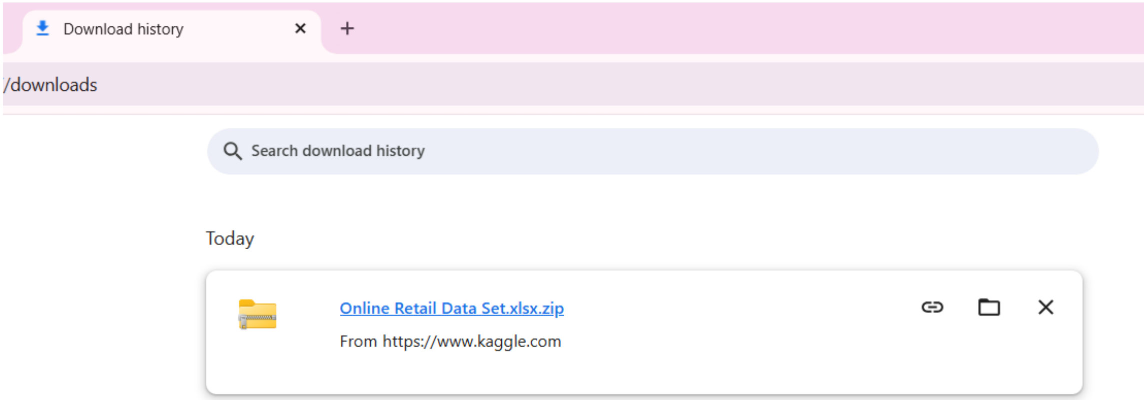


Figure 2: Screenshot of proof of download

Below is a screenshot of raw dataset view in Power BI

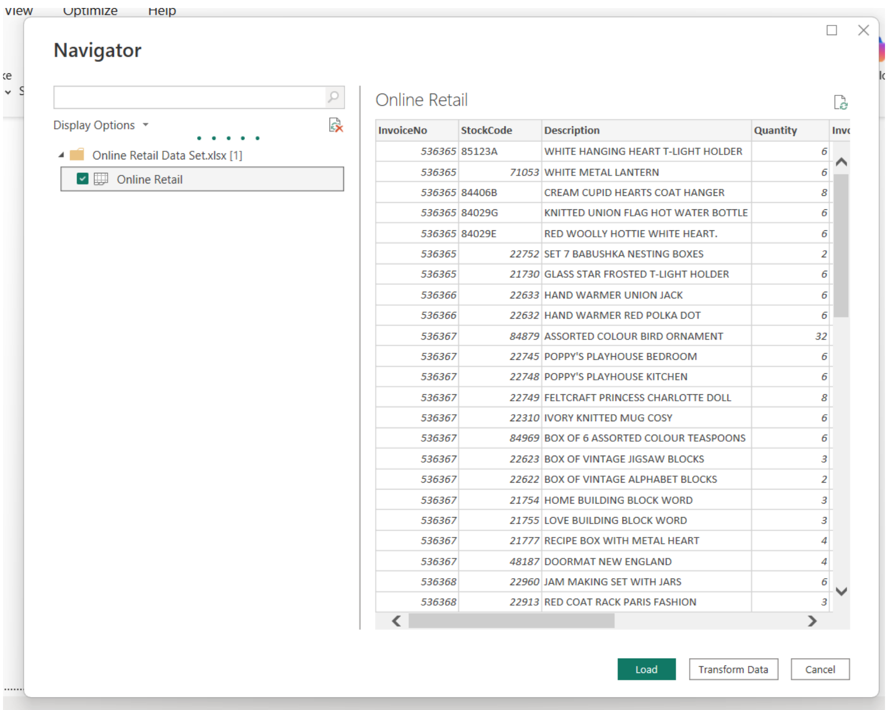


Figure 3: Screenshot of raw dataset view in Power BI

Below is a screenshot of successful load of dataset in Power BI

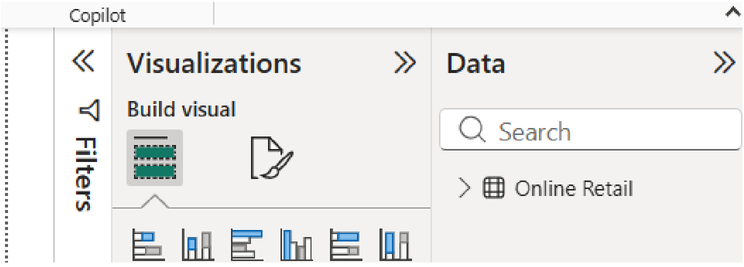


Figure 4: Screenshot of successful load of dataset in Power BI

3. Data Cleaning and Transformation in Power Query

3.1. Importing the Dataset into Power BI and opening in Power Query

Opening the dataset in Power Query allows systematic inspection, cleaning, and transformation before loading it into the data model, ensuring analytical accuracy and consistency.

Figure 5: Screenshot of successful import of dataset in Power Query

3.2. Inspecting Data Quality

Understanding data quality issues upfront helps justify later transformations. I checked for column data types, null values (CustomerID, Description), and negative Quantity values.

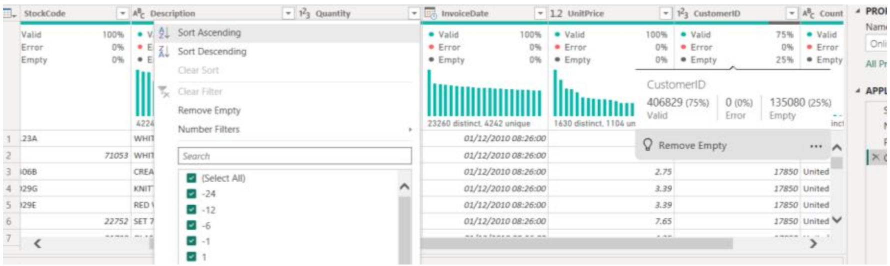


Figure 6: Screenshot of some of the data quality issues

3.3. Setting Correct Data Types

Setting correct data types ensures numerical accuracy, enables time-based analysis, and prevents DAX calculation errors in my later stages.

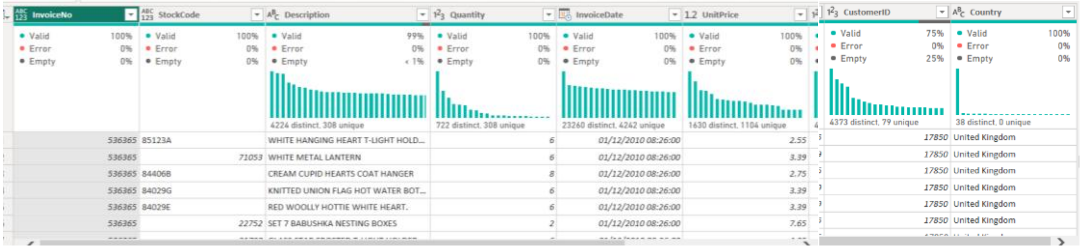


Figure 7: Screenshot of data before applying data type changes

Column Name	Correct Data Type Applied
InvoiceNo	Text
StockCode	Text
Description	Text
Quantity	Whole Number
UnitPrice	Decimal Number
InvoiceDate	Date/Time
CustomerID	Text
Country	Text

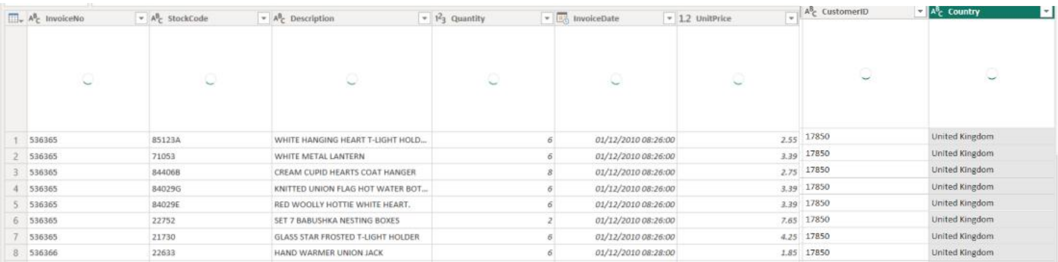


Figure 8: Screenshot of data after applying data type changes

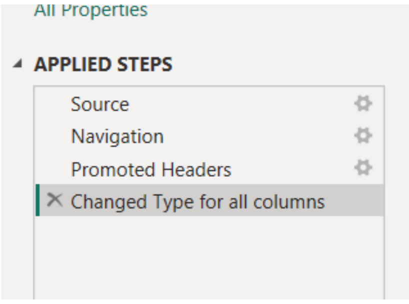


Figure 9: Screenshot of Applied Steps

3.4. Creating IsCancelled Column (for Cancelled Invoices)

I created an IsCancelled flag column to identify cancelled invoices. In the dataset, cancelled transactions are denoted by invoice number beginning with letter "C". Adding this flag allows greater analytical flexibility by enabling separate analysis of returns without data loss.

If 'IsCancelled' = Yes; then Cancelled invoice

If 'IsCancelled' = No; then Completed Sale

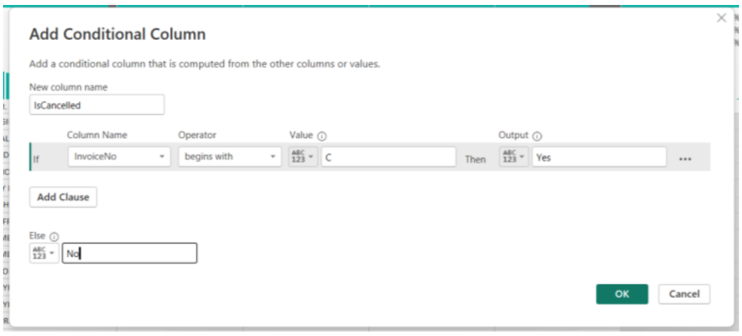
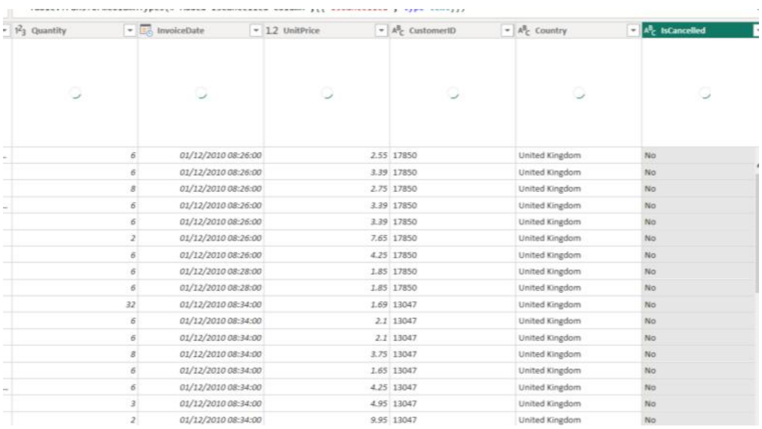


Figure 10: Screenshot of conditional column setup window



Quantity	InvoiceDate	UnitPrice	CustomerID	Country	IsCancelled
6	01/12/2010 08:26:00	2.55	17850	United Kingdom	No
6	01/12/2010 08:26:00	3.39	17850	United Kingdom	No
8	01/12/2010 08:26:00	2.75	17850	United Kingdom	No
6	01/12/2010 08:26:00	3.39	17850	United Kingdom	No
6	01/12/2010 08:26:00	3.39	17850	United Kingdom	No
2	01/12/2010 08:26:00	7.65	17850	United Kingdom	No
6	01/12/2010 08:26:00	4.25	17850	United Kingdom	No
6	01/12/2010 08:28:00	1.85	17850	United Kingdom	No
6	01/12/2010 08:28:00	1.85	17850	United Kingdom	No
32	01/12/2010 08:34:00	1.69	13047	United Kingdom	No
6	01/12/2010 08:34:00	2.1	13047	United Kingdom	No
6	01/12/2010 08:34:00	2.1	13047	United Kingdom	No
8	01/12/2010 08:34:00	3.75	13047	United Kingdom	No
6	01/12/2010 08:34:00	1.65	13047	United Kingdom	No
6	01/12/2010 08:34:00	4.25	13047	United Kingdom	No
3	01/12/2010 08:34:00	4.95	13047	United Kingdom	No
2	01/12/2010 08:34:00	9.95	13047	United Kingdom	No

Figure 11: Screenshot of dataset preview with IsCancelled column visible

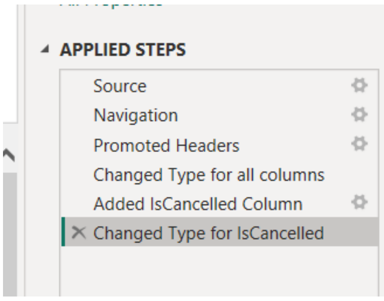


Figure 12: Screenshot of Applied Steps pane showing the new step

3.5. Returns Handling (Keeping negative quantities)

As seen during data inspection, ‘Quantity’ column has negative values. I decided to retain them because negative quantities indicate product returns. Retaining them ensures net revenue reflects actual business performance.

3.6. Creating Revenue Column

I created a revenue column to improve measure performance, and ensure consistent revenue calculations across visuals.

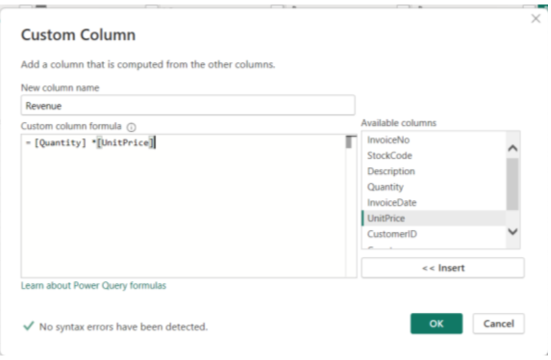


Figure 13: Screenshot of custom column formula editor

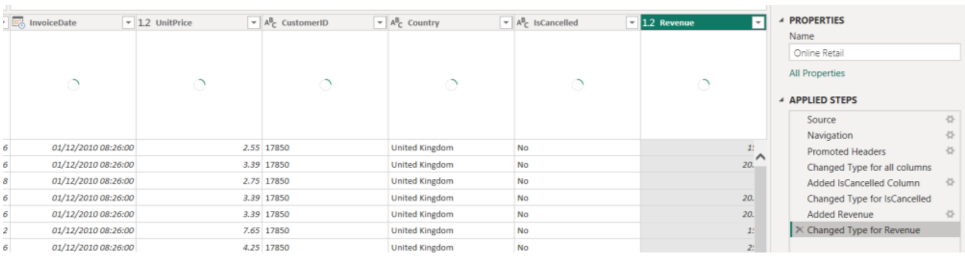


Figure 14: Screenshot of Revenue column's visibility

3.7. Removing Invalid Records (Zero or Negative Prices)

I removed records where UnitPrice is less than or equal to 0, as these do not represent valid product sales. These could be present in the dataset due to data entry errors, or as promotional placeholders, but since I'm not fully sure, I decided to remove them. Retaining such values would distort revenue calculations, average price metrics, and overall business performance analysis. Filtering these values ensures data duality and ensures analytical accuracy.

Figure 15: Screenshot of UnitPrice filter dialog box

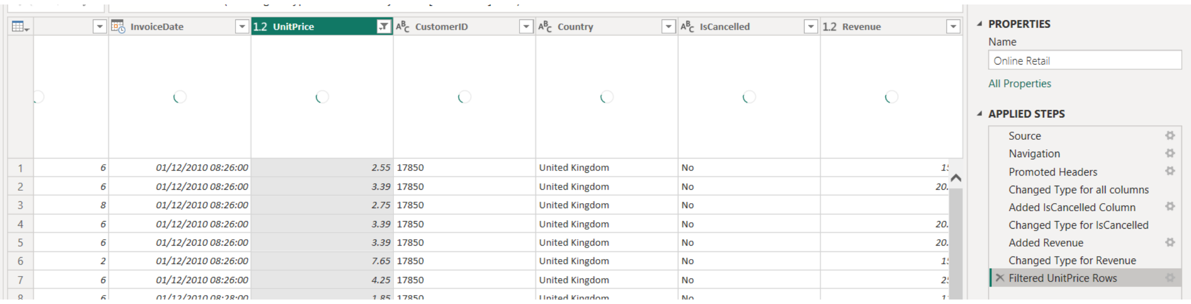


Figure 16: Screenshot of Dataset preview after filtering and Applied Steps pane with the filter step visible

3.8. Handling Missing CustomerID (Replacing with 'Unknown')

I replaced missing customer identifiers with "Unknown" to preserve valid sales transactions that lacked customer attribution. This ensures total sales and product performance metrics would remain accurate, while allowing customer-level analyses to exclude unidentified customers.

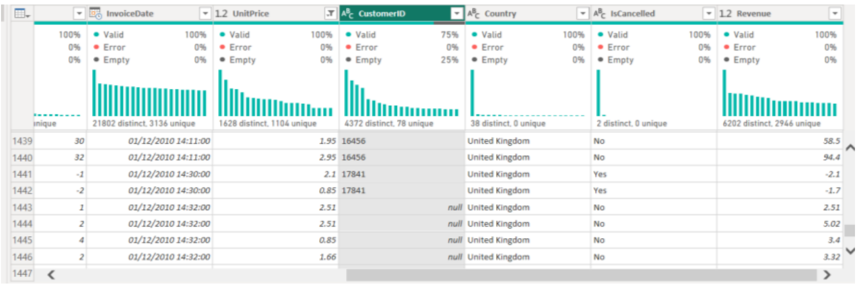


Figure 17: Screenshot of CustomerID column before replacement (showing nulls)

Figure 18: Screenshot of Replace Values dialog

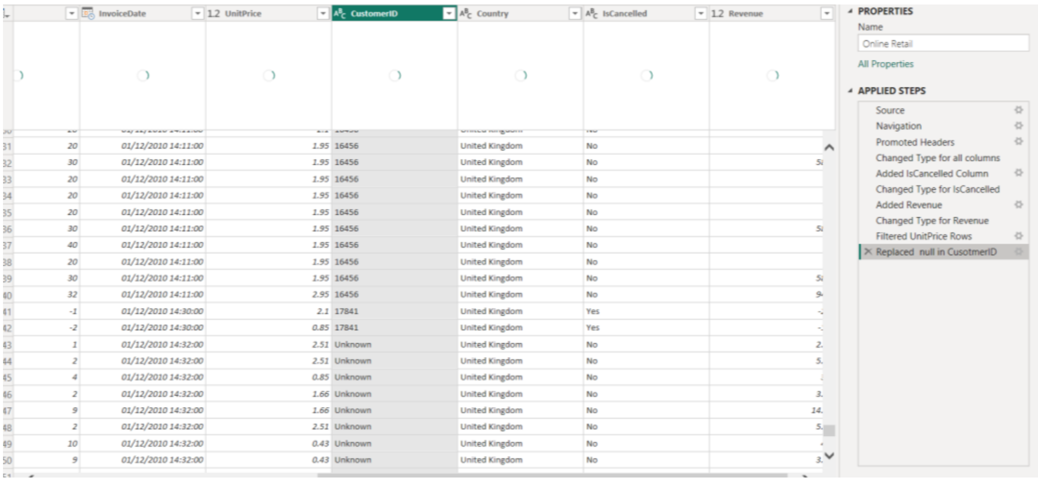


Figure 19: Dataset preview after replacement and Applied Steps showing Replace Values step

3.9. Extracting Date features from InvoiceDate

I extracted date parts from InvoiceDate which enables time-based grouping, trend analysis, and relationships with a Date Table.

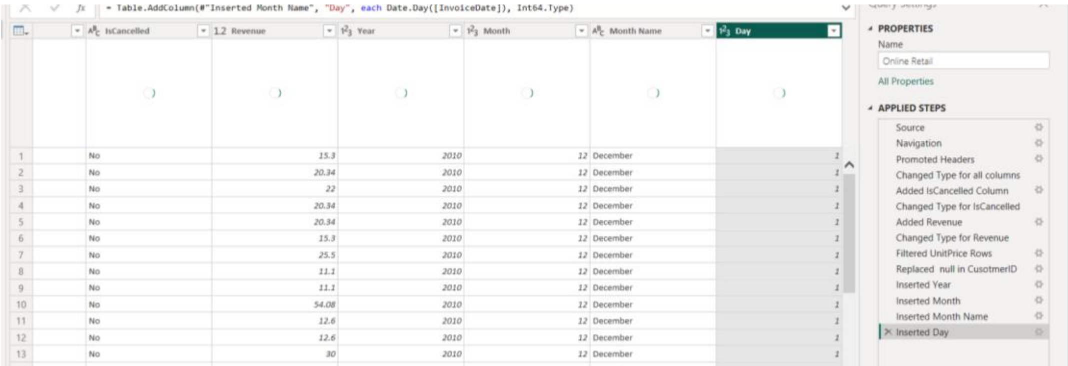


Figure 20: Screenshot of new date columns and Applied Steps pane

3.10. Renamed and cleaned columns

Clear, business-friendly column names and test cleaning improve model readability and make dashboards easier for non-technical stakeholders to interpret.

I renamed:

- InvoiceNo to Invoice No
- StockCode to Product Code
- Description to Product Name
- InvoiceDate to Transaction Date
- UnitPrice to Unit Price

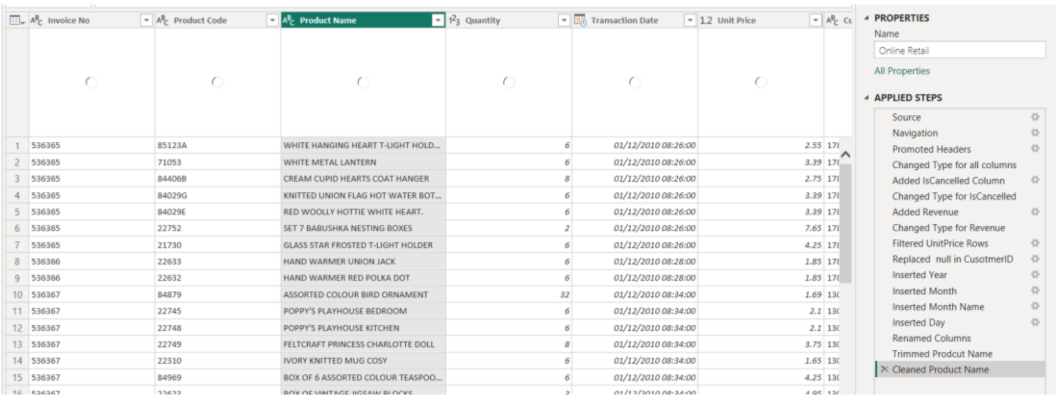


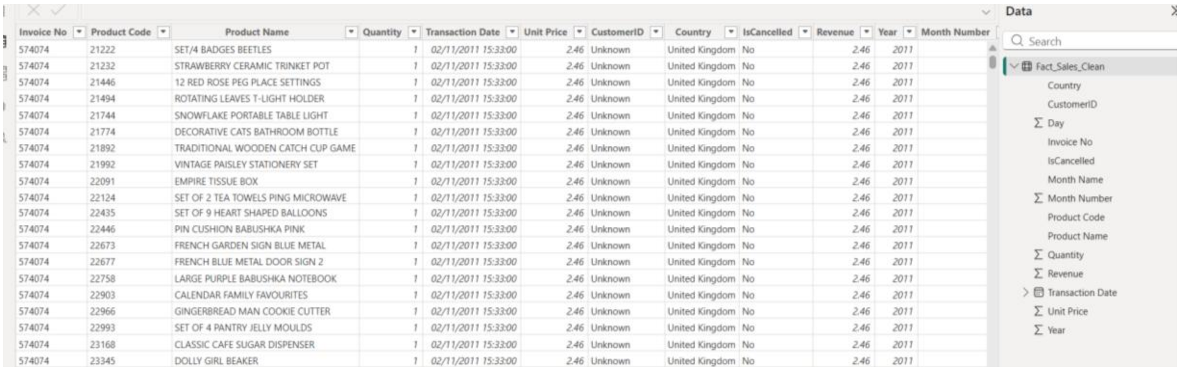
Figure 21: Screenshot of Columns renamed and Applied Steps pane

In summary, I used Power Query to clean and transform raw transaction data to ensure analytical accuracy and consistency. Key steps included correcting data types, creating a conditional column (IsCancelled), keeping negative quantities, creating Revenue column, removing invalid records, handling missing customer identifiers, renaming and cleaning columns, and extracting date features to support time-based analysis.

4. Data Modeling

4.1. Identifying the Fact Table

My cleaned dataset (Fact_Sales_Clean) represents transaction-level sales; thus, it is my fact table. This is because it contains quantitative measures such as Quantity, Revenue, Unit Price, and references dimension tables using keys, making it the central fact table.

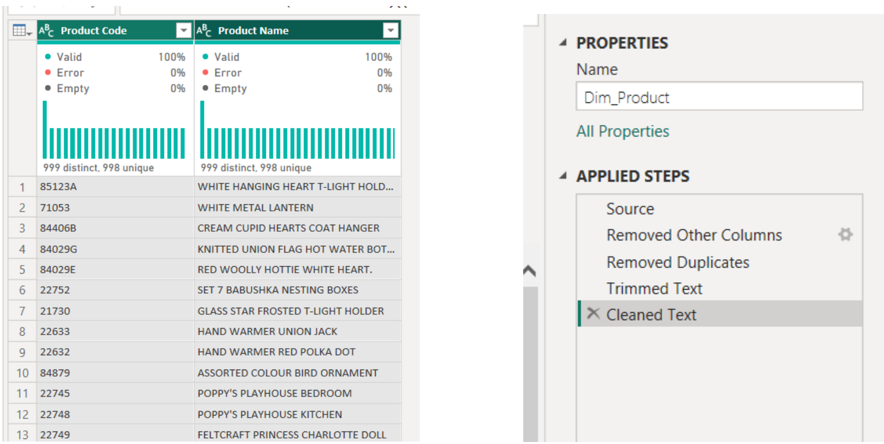


Invoice No	Product Code	Product Name	Quantity	Transaction Date	Unit Price	CustomerID	Country	IsCancelled	Revenue	Year	Month Number
574074	21222	SET/4 BADGES BEETLES	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	21232	STRAWBERRY CERAMIC TRINKET POT	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	21446	12 RED ROSE PEG PLACE SETTINGS	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	21494	ROTATING LEAVES T-LIGHT HOLDER	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	21744	SNOWFLAKE PORTABLE TABLE LIGHT	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	21774	DECORATIVE CATS BATHROOM BOTTLE	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	21892	TRADITIONAL WOODEN CATCH CUP GAME	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	21992	VINTAGE PAISLEY STATIONERY SET	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	22091	EMPIRE TISSUE BOX	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	22124	SET OF 2 TEA TOWELS PING MICROWAVE	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	22435	SET OF 9 HEART SHAPED BALLOONS	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	22446	PIN CUSHION BABUSHKA PINK	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	22673	FRENCH GARDEN SIGN BLUE METAL	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	22677	FRENCH BLUE METAL DOOR SIGN 2	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	22758	LARGE PURPLE BABUSHKA NOTEBOOK	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	22903	CALENDAR FAMILY FAVOURITES	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	22966	GINGERBREAD MAN COOKIE CUTTER	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	22993	SET OF 4 PANTRY JELLY MOULDS	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	23168	CLASSIC CAFE SUGAR DISPENSER	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	
574074	23345	DOLLY GIRL BEAKER	1	02/11/2011 15:33:00	2.46	Unknown	United Kingdom	No	2.46	2011	

Figure 22: Screenshot of data view showing the cleaned sales table with clearly visible column list

4.2. Dimension Table: Product Dimension

Products describe what is being sold and provide attributes used for grouping, filtering, and ranking. In addition, separating product attributes reduces redundancy, improves storage efficiency, and supports scalable analysis.



	Product Code	Product Name
	Valid 100% / Error 0% / Empty 0% / 999 distinct, 998 unique	Valid 100% / Error 0% / Empty 0% / 999 distinct, 998 unique
1	85123A	WHITE HANGING HEART T-LIGHT HOLD...
2	71053	WHITE METAL LANTERN
3	84406B	CREAM CUPID HEARTS COAT HANGER
4	84029G	KNITTED UNION FLAG HOT WATER BOT...
5	84029E	RED WOOLLY HOTTIE WHITE HEART.
6	22752	SET 7 BABUSHKA NESTING BOXES
7	21730	GLASS STAR FROSTED T-LIGHT HOLDER
8	22633	HAND WARMER UNION JACK
9	22632	HAND WARMER RED POLKA DOT
10	84879	ASSORTED COLOUR BIRD ORNAMENT
11	22745	POPPY'S PLAYHOUSE BEDROOM
12	22748	POPPY'S PLAYHOUSE KITCHEN
13	22749	FELTCRAFT PRINCESS CHARLOTTE DOLL

Figure 23: Screenshot of data preview of Dim_Product Table and its Applied Steps

4.3. Dimension Table: Customer Dimension

Customer attributes are isolated to enable customer-level aggregation and segmentation without duplicating data across transactions.

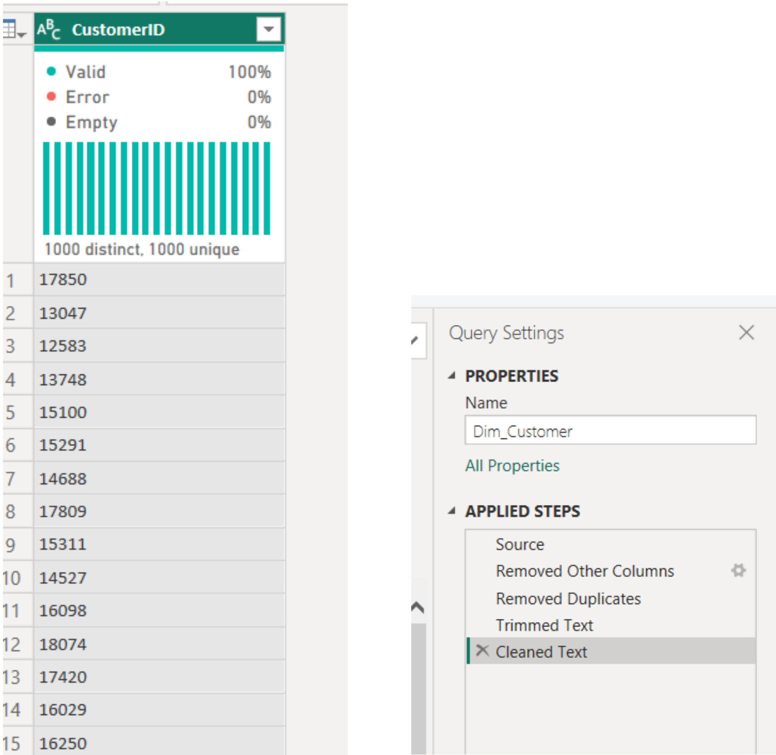


Figure 24: Screenshot of data preview of Dim_Customer Table and its Applied Steps

4.4. Dimension Table: Country Dimension

I created a separate country dimension to support geographic slicing and avoid repeating country values in the fact table.

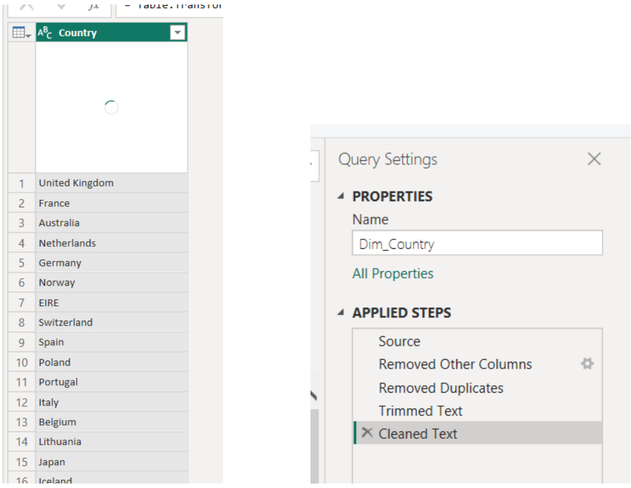


Figure 25: Screenshot of data preview of Dim_Country Table and its Applied Steps

4.5. Dimension Table: Date Dimension

The Date table enables advance time-intelligence calculations such as year-to-date, month-to-date, and trend analysis. Using a dedicated Date table ensures consistency and improves model performance.

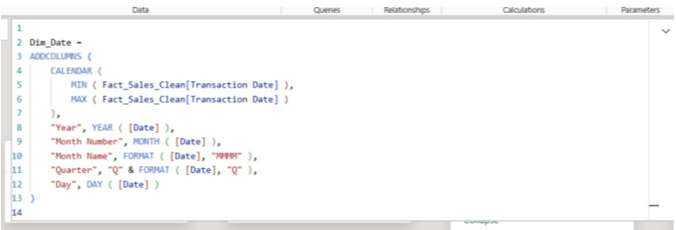


Figure 26: Screenshot of DAX formula for Date table

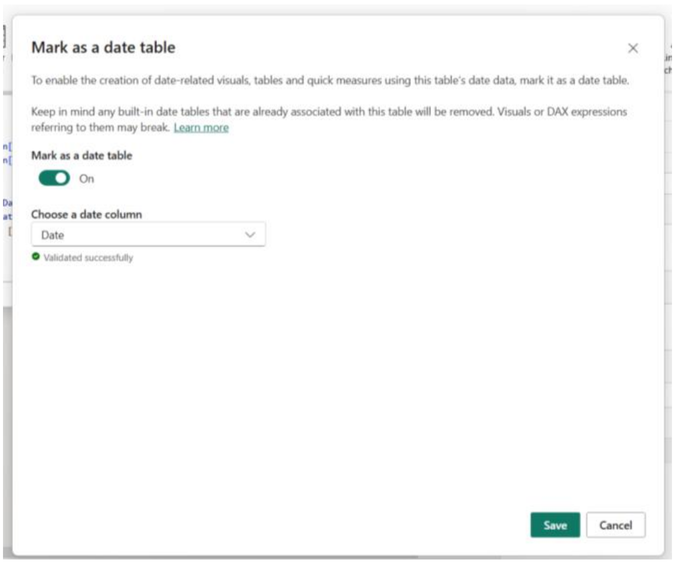
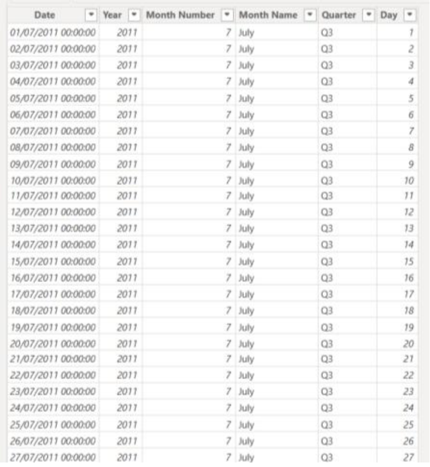


Figure 27: Screenshot of marking it as a date table



Date	Year	Month Number	Month Name	Quarter	Day
01/07/2011 00:00:00	2011	7	July	Q3	1
02/07/2011 00:00:00	2011	7	July	Q3	2
03/07/2011 00:00:00	2011	7	July	Q3	3
04/07/2011 00:00:00	2011	7	July	Q3	4
05/07/2011 00:00:00	2011	7	July	Q3	5
06/07/2011 00:00:00	2011	7	July	Q3	6
07/07/2011 00:00:00	2011	7	July	Q3	7
08/07/2011 00:00:00	2011	7	July	Q3	8
09/07/2011 00:00:00	2011	7	July	Q3	9
10/07/2011 00:00:00	2011	7	July	Q3	10
11/07/2011 00:00:00	2011	7	July	Q3	11
12/07/2011 00:00:00	2011	7	July	Q3	12
13/07/2011 00:00:00	2011	7	July	Q3	13
14/07/2011 00:00:00	2011	7	July	Q3	14
15/07/2011 00:00:00	2011	7	July	Q3	15
16/07/2011 00:00:00	2011	7	July	Q3	16
17/07/2011 00:00:00	2011	7	July	Q3	17
18/07/2011 00:00:00	2011	7	July	Q3	18
19/07/2011 00:00:00	2011	7	July	Q3	19
20/07/2011 00:00:00	2011	7	July	Q3	20
21/07/2011 00:00:00	2011	7	July	Q3	21
22/07/2011 00:00:00	2011	7	July	Q3	22
23/07/2011 00:00:00	2011	7	July	Q3	23
24/07/2011 00:00:00	2011	7	July	Q3	24
25/07/2011 00:00:00	2011	7	July	Q3	25
26/07/2011 00:00:00	2011	7	July	Q3	26
27/07/2011 00:00:00	2011	7	July	Q3	27

Figure 28: Screenshot of data view of Dim_Date

4.6. Creating Relationships between tables

Relationship Summary

From (Fact)	To (Dimension)	Cardinality	Filter Direction
Product Code	Dim_Product[Product Code]	Many → One	Single
CustomerID	Dim_Customer[CustomerID]	Many → One	Single
Country	Dim_Country[Country]	Many → One	Single
Transaction Date	Dim_Date[Date]	Many → One	Single

Single-direction filtering from dimension to fact ensures predictable behavior, prevents ambiguity, and optimizes performance in a star schema.

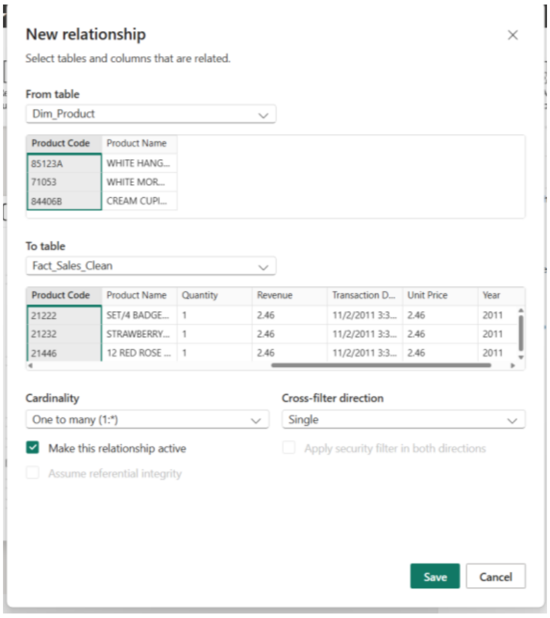


Figure 29: Screenshot of relationship between Dim_Product[Product Code] and Fact_Sales_Clean[Product Code]

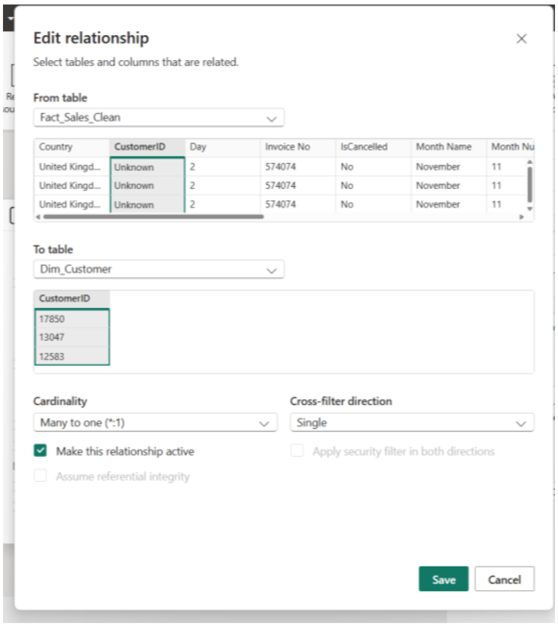


Figure 30: Screenshot of relationship between Dim_Customer[CustomerID] and Fact_Sales_Clean[CustomerID]

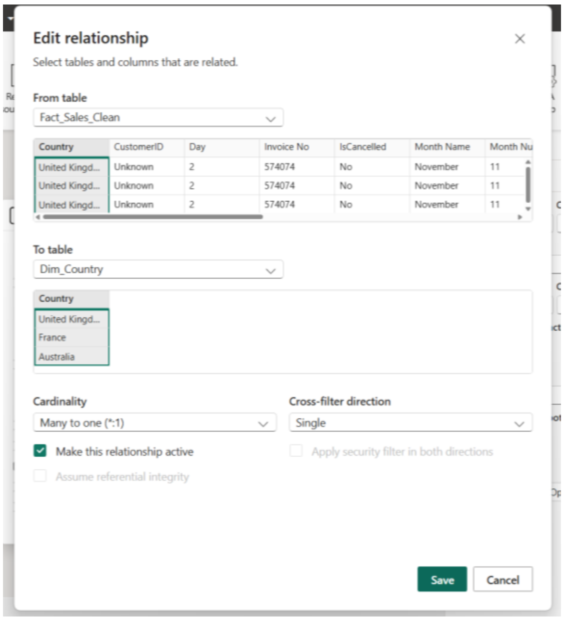


Figure 31: Screenshot of relationship between Dim_Country[Country] and Fact_Sales_Clean[Country]

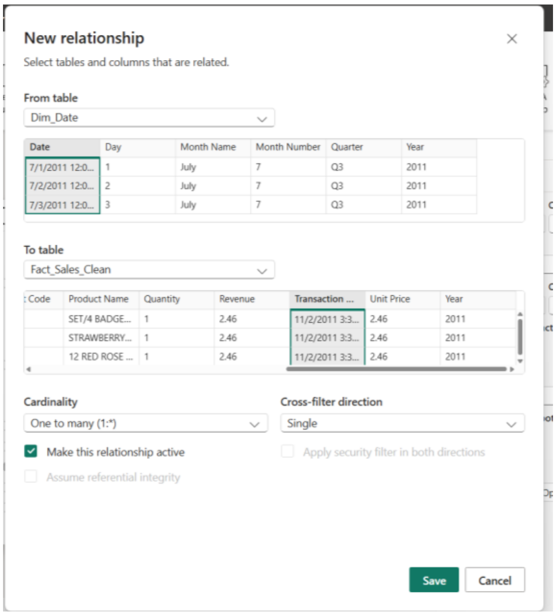


Figure 32: Screenshot of relationship between Dim_Date[Date] and Fact_Sales_Clean[Transaction Date]

4.7. Model View

The data model follows a star schema design consisting of one central fact table containing sales transactions and multiple dimension tables representing customers, products, countries, and dates. This star schema improves reporting performance by minimizing relationships and simplifying filter propagation. Separating dimensions such as Product, Customer, and Date ensures clean aggregation logic and enables scalable DAX measures for analytical reporting.

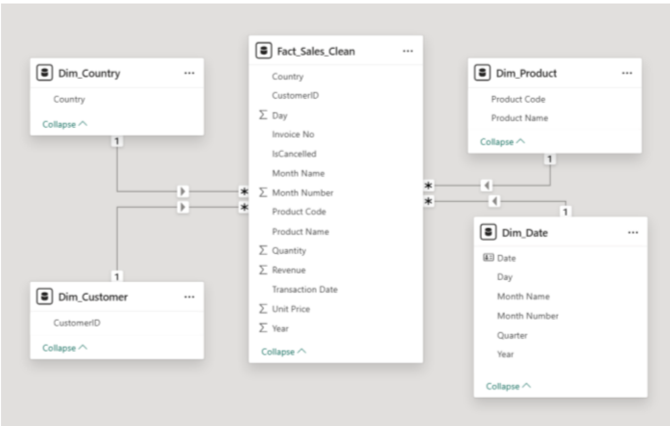


Figure 33: Screenshot of model view showing all tables and relationships

5. DAX Measures and Calculated Columns

5.1. Calculated columns

Calculated columns are row-level attributes stored in the model and used for grouping, filtering, and slicing.

SalesAmount: The SalesAmount column represents the monetary value of each transaction line and is used as the base metric for revenue calculations.

```
SalesAmount =
Fact_Sales_Clean[Quantity] * Fact_Sales_Clean[Unit Price]
```

Figure 34: Screenshot of SalesAmount DAX formula

Transaction Type: This column classifies each transaction as completed or cancelled, enabling separation of valid sales from cancellations during analysis.

```
Transaction Type =
IF (
    Fact_Sales_Clean[IsCancelled] = "Yes" || Fact_Sales_Clean[Quantity] < 0,
    "Return (Cancelled)",
    "Sale (Completed)"
)
```

Figure 35: Screenshot of Transaction Type DAX formula

Revenue Band: Revenue bands help segment transactions by value, enabling identification of high-impact sales events.

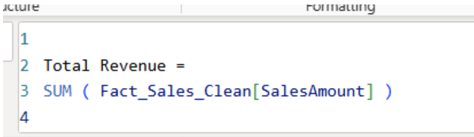
```
Revenue Band =
IF (
    Fact_Sales_Clean[Revenue] < 50,
    "Low",
    IF (
        Fact_Sales_Clean[Revenue] < 200,
        "Medium",
        "High"
    )
)
```

Figure 36: Screenshot of Revenue Band DAX formula

5.2. DAX Measures

Measures are dynamic calculations evaluated at query time and respond to filters and slicers.

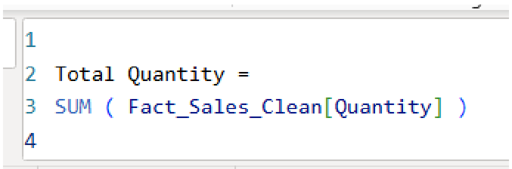
Total Revenue: Calculates total sales revenue across all selected dimensions.



```
Total Revenue =
SUM ( Fact_Sales_Clean[SalesAmount] )
```

Figure 37: Screenshot of Total Revenue DAX formula

Total Quantity: Measures overall sales volume and demand intensity.



```
Total Quantity =
SUM ( Fact_Sales_Clean[Quantity] )
```

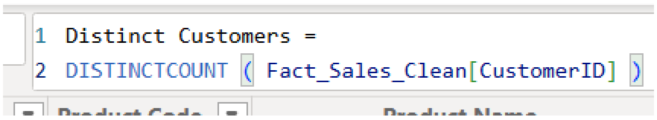
Figure 38: Screenshot of Total Quantity DAX formula

Total Transactions: Shows transaction volume independent of quantity and price effects.

```
Total Transactions =
DISTINCTCOUNT ( Fact_Sales_Clean[Invoice No] )
```


Figure 39: Screenshot of Total Transactions DAX formula

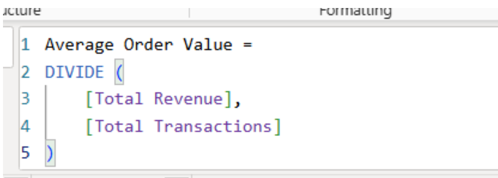
Distinct Customers: Indicates customer reach and engagement.



```
Distinct Customers =
DISTINCTCOUNT ( Fact_Sales_Clean[CustomerID] )
```

Figure 40: Screenshot of Distinct Customers DAX formula

Average Order Value: Shows the average revenue generated per transaction which helps evaluate pricing strategy and customer purchasing power.



```
Average Order Value =
DIVIDE (
    [Total Revenue],
    [Total Transactions]
)
```

Figure 41: Screenshot of Average Order Value DAX formula

Revenue % Contribution by Product: Shows how much each product contributes to total revenue

```
Revenue % of Total =
DIVIDE (
    [Total Revenue],
    CALCULATE ( [Total Revenue], ALL ( Dim_Product ) )
)
```

Figure 42: Screenshot of Revenue % of Total DAX formula

Year-to-Date Revenue (YTD) (Time Intelligence): Allows cumulative performance tracking across time.

```
Revenue YTD =
TOTALYTD (
    [Total Revenue],
    Dim_Date[Date]
)
```

Figure 43: Screenshot of Revenue YTD DAX formula

Previous Year Revenue: Used for growth analysis. It enables historical comparison and trend diagnosis

```
Previous Year Revenue =
CALCULATE (
    [Total Revenue],
    SAMEPERIODLASTYEAR ( Dim_Date[Date] )
)
```

Figure 44: Screenshot of Previous Year Revenue DAX formula

Year-on-Year Growth %: Measures growth or decline which is critical for executive decision-making.

```

YoY Growth % =
DIVIDE (
    [Total Revenue] - [Previous Year Revenue],
    [Previous Year Revenue]
)
```

Figure 45: Screenshot of YoY Growth % DAX formula

In summary, a set of calculated columns and DAX measures were created to support dynamic and interactive analysis. Calculated columns were used for transaction-level classification and revenue computation, while measures were designed to compute key performance indicators such as total revenue, average order value, customer count, revenue contribution, time-based trends, growth metrics, and product ranking. These calculations enable robust business analysis.

6. Dashboard Design and Visualization

6.1. Executive Summary

KPI cards provide at-a-glance insight into overall business performance for senior management.

Line charts are ideal for time-series analysis and help identify growth patterns, trends, seasonality, or anomalies.

Clustered bar chart allows management to identify top-performing geographic markets.

Slicers enable user-driven exploration and increase dashboard interactivity.

Figure 46: Screenshot of Executive Summary dashboard page

6.2. Detailed Analysis

In clustered bar chat drill-down enables hierarchical exploration without cluttering the dashboard.

A matrix supports detailed comparison across products and time, ideal for operational analysis.

Donut chart identifies dependency on top-performing products and revenue concentration risk.

A decomposition tree was used to perform root-cause analysis of total revenue, allowing users to interactively drill down from overall performance to country-level, product-level, and time-based contributors. These visual supports exploratory analysis and helps identify the primary drivers of revenue changes across dimensions.

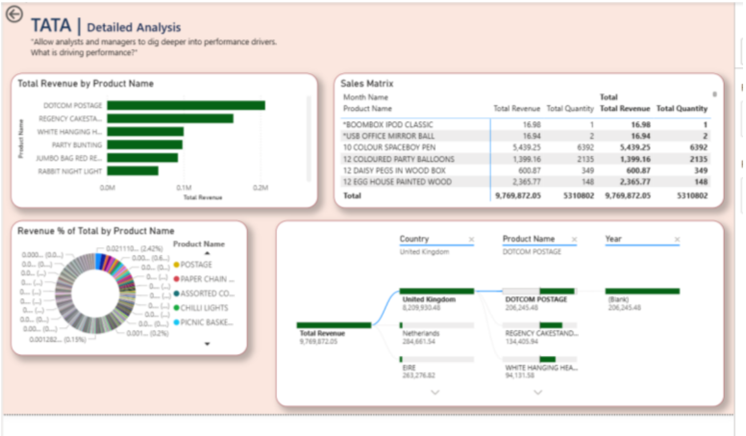


Figure 47: Screenshot of Detailed Analysis dashboard page

6.3. Insights & Performance Monitoring

Line chart supports year-over-year performance evaluation.

A donut chart helps evaluate operational efficiency and sales quality.

Clustered bar chart highlights best performers and underperformers for immediate action.

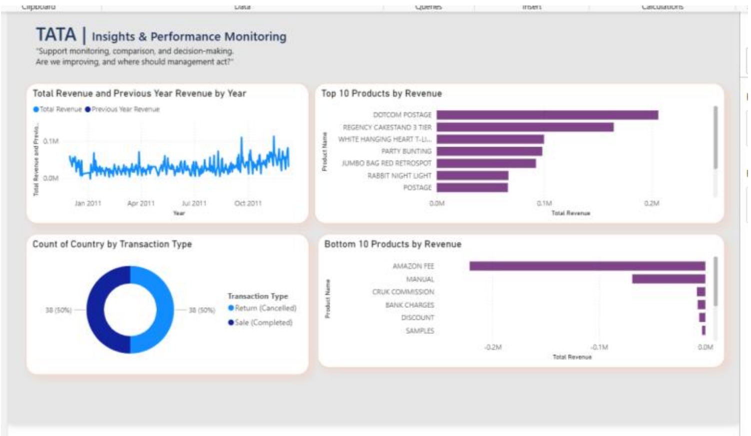


Figure 48: Screenshot of Insight & Performance Monitoring

In summary, the dashboard was designed using a three-page structure to support executive oversight, detailed analytical exploration, and performance monitoring. Page one provides a high-level summary of key metrics and trends for senior stakeholders. Page two enables deeper analysis through drill-down, ranking, and decomposition visuals to identify performance drivers. Page three focuses on time-based comparison and monitoring, enabling identification of growth patterns, risks, and operational issues. Interactive slicers and cross-filtering were implemented to enhance user-driven exploration.

7. Analytical Insights and Business Recommendations

In this section, I aim to interpret the Power BI dashboard to extract meaningful business insights and translate them into actionable recommendations.

From the Executive Summary page, the Revenue by Country bar chart shows that the United Kingdom contributes the vast majority of total revenue, significantly outweighing all other countries. This suggests that business is heavily dependent on one market, which exposes it to geographic concentration risk.

The Revenue Trend line chart indicates a general upward trend over time, but with frequent short-term spikes and dips, especially toward the later periods. This suggests that while revenue is growing, demand appears uneven, suggesting seasonality, promotional effects, or operational fluctuations impacting sales consistency.

The Top 10 Products by Revenue bar chart in both the Detailed Analysis and Insights & Performance Monitoring pages shows that a small subset of products accounts for a large portion of total revenue. At the same time, the Bottom 10 Products by Revenue chart highlights multiple products contributing minimal or negative value. This suggests that product performance follows the Pareto (80/20) principle, where few products generate most returns while many contribute little.

The decomposition tree on the Detailed Analysis page reveasl that top-performing products differ across countries, with some products performing strongly in the UK but weakly in other regions. This suggests that customer preferences are geographically distinct.

The Transaction Type visual shows that completed sales dominate total transactions, but a noticeable portion of cancelled transactions still exists. This suggests that operational issues such as returns, order errors, or customer dissatisfaction could be causing avoidable revenue leakage.

Recommendations include:

Expanding promotion and visibility of mid-tier products rather than relying only on top sellers. This is because diversifying revenue sources reduces operational risk and stabilizes long-term performance.

Aligning inventory procurement and marketing campaigns with peak sales periods identified in the revenue trend. This is because preparing for demand spikes minimizes stock-outs and maximizes revenue capture during high-demand periods.

Analyzing high-return products to identify common causes (e.g., pricing errors, misleading descriptions, quality issues). This is because reducing return rates improves net revenue and customer satisfaction.

Targeting high-potential non-UK markets through focused product offerings or localized promotional strategies. This is because improving revenue contribution from international markets reduces dependency on a single country and supports growth.

8. Conclusion

The analysis reveals strong overall performance driven by a limited set of products and markets, alongside clear opportunities for diversification and operational improvement. By expanding geographically, optimizing the product portfolio, and addressing cancellation drivers, the business can enhance revenue stability and long-term profitability.

9. GitHub repository link

Below is the link to the GitHub Repository:

https://github.com/HetalK4/TATA-Online-Retail-Dataset